Homework 6

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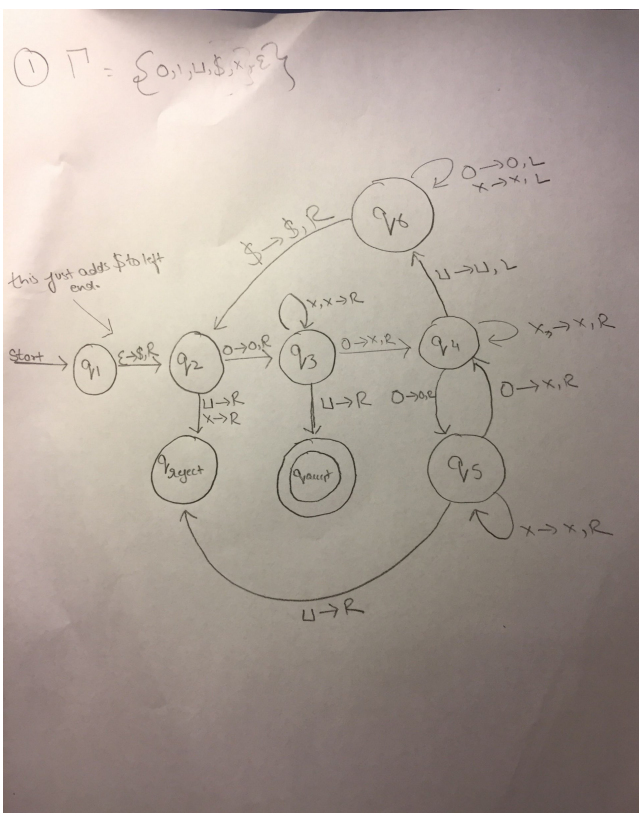
May 13, 2018

Problem 1. In class we discussed the language A=02n:n>=0. We looked in detail at the TM given in that text, that accepts A. That TM starts by converting the first 0 to a space, and uses that space to recognize when the TM is at the left end of the tape. But, since that 0 is not x-ed out,the number of un-xed 0s after a sweep is one more than the number of 0s literally on the tape. This off-by-one seemed to make it harder to understand the correctness of the TM. Some students suggested not converting the first 0 to a space. So, let assume that the TM has the ability to detect when it has hit the left end of the tape, i.e. the read/write head is on square one, so it cannot move further left. With that assumption, work out the details for a TM to accept language A without converting the first 0 to a space. Draw out the state diagram for this TM.

Proof.



□

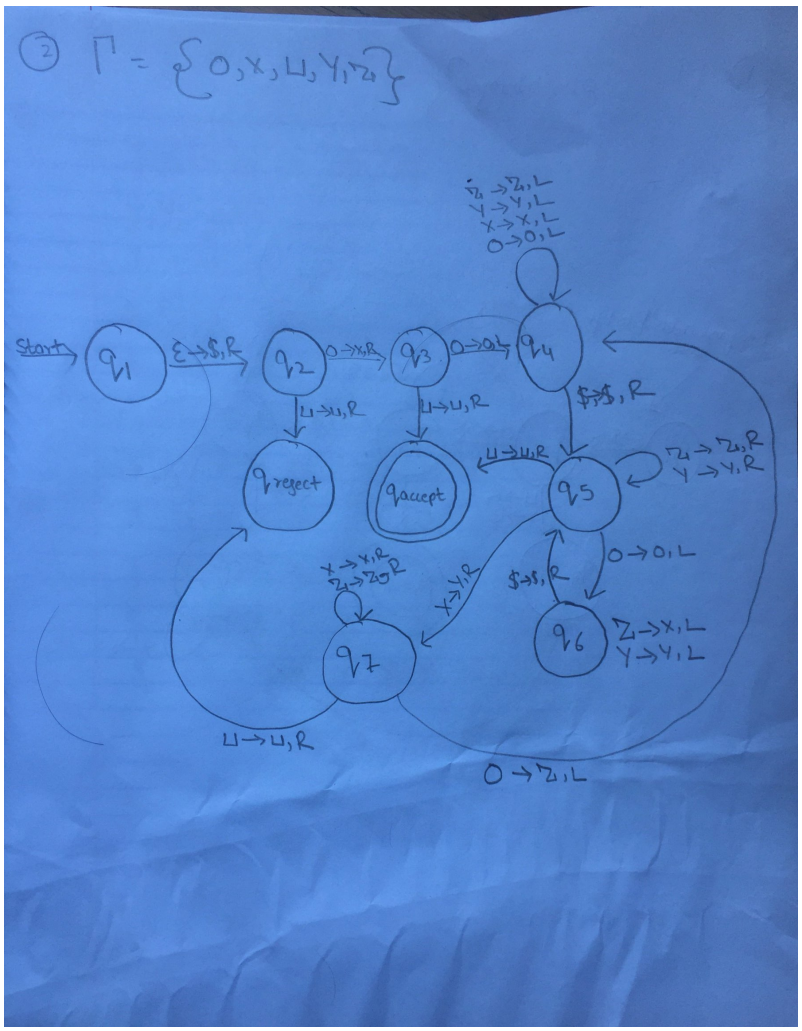
Problem 2. Another student suggested a different algorithm to accept language A. The idea is to again make several sweeps of the tape, and again x-out a number of 0s in each sweep. In detail, the first sweep xs one 0. The second sweep xs another 0. The third sweep xs out two more 0s. Each successive sweep xs out a number of 0s equal to the total number of 0s that have been xd out by the end of the prior sweep. The consequence is that the total number of xs on the tape doubles in each sweep.

Problem 2a. What is the end condition for this algorithm? That is, how does this algorithm finally decide if the input string is in A, or is not in A?

Proof. **Algorithm Outline**
1. The minimum length has to be at least one; therefore, check if you can x out at least one 0 and move right.
2. If step one is impossible, Reject.(END CONDITION 1)
3. If step one is successful, and after moving right the right end is detected i.e the string only has one 0, Accept (END CONDITION 2)
4. If 0 is detected after moving right on step one, then keep moving left till you detect left end.
5. Move right to find a x, replace it with a symbol y, and then move right till you find a 0 and replace the 0 with symbol z. Move left till left end is detected.
6. Repeat step 5, till no more x's are left.
7. If during step 5 or during step 6. You do not find a 0 for a x, then Reject. (END CONDITION 3)
8. If all x's are replaced by y, and you have z'd out a corresponding 0, then convert all y's and z's to x's.
9. Move right till you detect a right end or 0.
10. If right end is detected, Accept (END CONDITION 4).
11. If 0 is detected, move to left end and go to step 5.

□

Problem 2b. Now we turn to TM programming. Figure out, and draw out the state diagram with all of its details, to implement this algorithm. Explain the program. Remember that a TM does not have any built-in numbers or counting or arithmetic ability. So, how is the doubling of the xs implemented?
Hint 1: Zig-Zag.
Hint 2: You almost surely will need a write alphabet that contains extra characters beyond {0, x,t}.



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Problem 3.15. Show that the collection of decidable languages is closed under the operations of

Problem b Concatenation.

Proof. Let M1 and M2 be TMs that accept language L1 and L2.

For string w, divide w in y and z for every combination possible for left to right.(i.e for a string 0011 y= 11 and z = 00 is not possible)
Run M1 on y and M2 on z.
If both M1 and M2 accept, then accept, else reject.

□

Problem 3.15c star.

Proof. Let TM M1 be the machine that accepts language L1.
LET TM M' be the TM for **star** operation on M1.
For M'
For a string w, cut it into pieces w1,w2,w3...wn.
Find all such combinations of string w left to right.
Run all the pieces through M1
If for any of the combination, all the pieces of that combination are accepted by M1, then M' accepts. Else reject.
Let TM M2, be the machine that accepts empty string, and rejects anything else.
Run M2 for string w.
If M2 accepts. Then M' accepts. If M2 Rejects, check the operations on M1 done above to determine if it accepts or rejects.

□

Problem 3.15d Complementation.

Proof. For TM M1 that accepts language L1.
Let there be a TM M1' that accepts complement of language L1.
M1' is same as M1 but replace Accept and Reject with each other.

□

Problem 3.16. Show that the collection of Turing-Recognizable languages is closed under the operations of

Problem b Concatenation.

Proof. Let M1 and M2 be TMs that accept language L1 and L2.

For string w, divide w in y and z for every combination possible for left to right.(i.e for a string 0011 y= 11 and z = 00 is not possible)
Run M1 on y and M2 on z.
If both M1 and M2 accept, then Accept.
else if either of it halts and rejects, then Reject.
NOTE: If they can both run forever, in which case if either of them runs forever so will the machine run forever. □

Problem 3.15c star.

Proof. Let TM M1 be the machine that accepts language L1.
LET TM M' be the TM for **star** operation on M1.
For M'
For a string w, cut it into pieces w1,w2,w3...wn.
Find all such combinations of string w left to right.
Run all the pieces through M1
If for any of the combination, all the pieces of that combination are accepted by M1, then M' Accepts. Else if it halts and rejects then Reject.
Let TM M2, be the machine that accepts empty string, and rejects anything else.
Run M2 for string w.
If M2 accepts. Then M' accepts. If M2 Rejects, check the operations on M1 done above to determine if it accepts or rejects.
NOTE: It is possible for M' to run forever , if the computations run forever. Hence we only reject if it halts and rejects. □

Problem 3.15d Intersection.

Proof. Let M1 be the language that accepts L1.
Let M2 be the language that accepts L2.
For a string w.
Let M' be the intersection of M1 and M2.
Run string w on M1, and M2 simultaneously for one character at a time.
If one of the machine accepts, continue computing the rest of the string for the other machine.
If the other machine accepts then M' will accept.
If the other machine goes on forever, then so will M'.
If either of the machine halts and rejects, then M' will reject.
NOTE: If computation on both go on forever so will M'. If one of them rejects the machine will halt. If one of them accepts, and other goes on forever so will the machine.

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